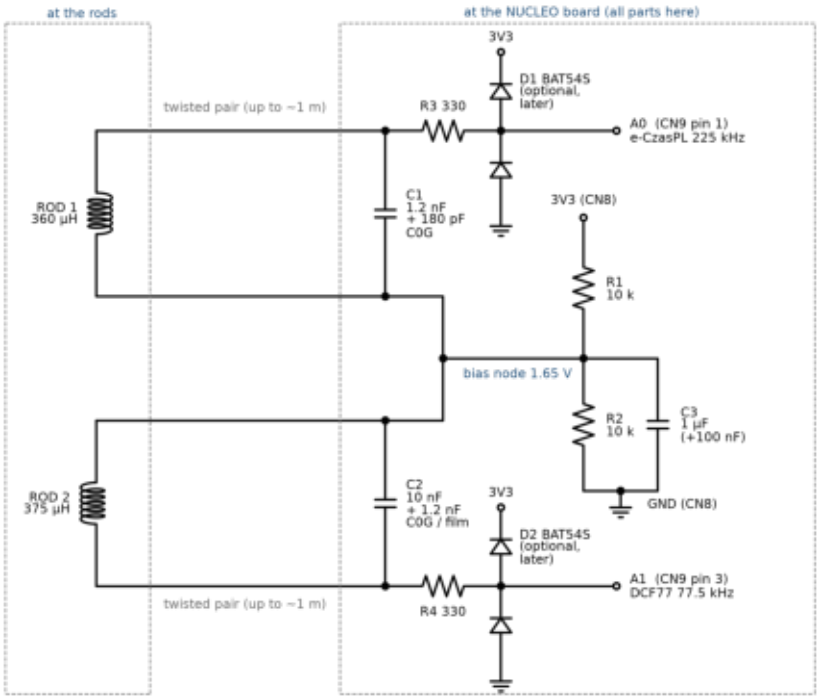
Two ferrite rods to NUCLEO-H723ZG: e-CzasPL on A0, DCF77 on A1

Rods alone at the antenna end; tuning capacitors and one shared 1.65 V bias node at the board



Parts

ROD 1: the rod on A0 today (12 cm, 360 µH); ROD 2: the second rod, measured 375 µH (bare coil)
C1 = 1.2 nF + 180 pF in parallel (1.38 nF) -> 225.8 kHz (today: 1.5 nF = 216.6 kHz)
C2 = 10 nF + 1.2 nF in parallel (11.2 nF) -> 77.7 kHz (ideal 11.25 nF; 10 nF + 1.5 nF would be 76.6 kHz)
C1, C2: C0G/NP0 ceramic, polystyrene or polypropylene film (stable, low loss; not X7R)
R1, R2 = 10 k; C3 = 1 µF ceramic, optionally + 100 nF, at the board next to R1 / R2
Protection per pin: R3, R4 = 330 Ohm in series (fit these now); D1, D2 = BAT54S or BAV99 optional, later

Connections

At the rods: nothing but the rod. Each rod -> its own twisted pair -> the board end.
At the board end: C1 across pair 1, C2 across pair 2 (parallel tuned circuits), R1, R2, C3.
Cold ends of both rods -> the bias node; 3V3 -> R1 -> bias node -> R2 -> GND; C3 bias node -> GND.
Hot end of C1 -> R3 330 Ohm -> A0 (CN9 pin 1); hot end of C2 -> R4 330 Ohm -> A1 (CN9 pin 3).
Later, if wanted: a dual diode at each pin (middle to the pin, ends to 3V3 and GND).
Put C1 / C2 right at the pins. The cable capacitance (about 50 pF per metre) adds to C1 / C2.

Why this is simpler and better

One divider and one decoupling capacitor serve both channels; the two 100 nF coupling capacitors are gone. The bias node is an AC ground (C3), so each tuned circuit sits between its pin and AC ground, and DC reaches the pin through the rod winding (a few ohms): both pins sit at 1.65 V.
Old input: the 10k/10k divider (5 k) loaded the tuned rod through 100 nF: Q about 8 instead of about 50, about 15 dB less signal at 225 kHz. Now nothing loads it (ADC sampling at 1 Msps is about 200 k).
C1 / C2 are also the charge reservoir for the ADC's sampling capacitor (a few pF, through 330 Ohm).
Protection: C1 / C2 already absorb fast spikes (AC-grounded by C3); 330 Ohm limits the current and the diodes clamp the pin to -0.3...3.6 V (STM32 pins: max 5 mA). Without the diodes, 330 Ohm alone helps.

Tuning (higher Q = narrower: 225 kHz about +-2 kHz, 77.5 kHz about +-0.8 kHz)

Rods vary by several %. Tune for the highest C/N0 in the NUCLEO's CARRIER line (every 60 s): add or remove small capacitors at the board (47-220 pF at 225 kHz, 0.47-1 nF at 77.5 kHz).
With 1 m of twisted pair (about 50 pF) C1 = 1.2 nF + 150 pF is closer; 77.5 kHz hardly changes.

Placement

A ferrite rod receives best broadside (its axis at right angles to the station's direction).
From Wroclaw: Solec Kujawski (225 kHz) is about north -> ROD 1 axis east-west;
Mainflingen (DCF77) is about west -> ROD 2 axis north-south. The rods end up crossed, which also keeps them from coupling; keep them 20-30 cm apart and away from the board and power supplies.

Firmware

Today both stations come from A0. A1 needs a firmware change (second ADC, DCF77 from A1); I make it after the rod is connected. Until then A1 is simply unused.